Transactional Longitudinal Relations Between Accuracy and Reaction Time on a Measure of Cognitive Flexibility at 5, 6, and 7 Years of Age

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Research Highlights

- Longitudinal associations between accuracy and reaction time on the DCCS at 5, 6, and 7 years were studied to examine developmental change in flexibility.
- Slowing down at 5 and again at 6 years preceded improved accuracy, which then predicted increased efficiency (speed) at 7 years.
- Working memory acted as a partial mediator between slower responding at 5 years and improved accuracy by 6 years, as suggested by the Iterative Reprocessing model.

Abstract

Whereas accuracy is used as an indicator of cognitive flexibility in preschool-age children, reaction time (RT), or a combination of accuracy and RT, provide better indices of performance as children transition to school. Theoretical models and cross-sectional studies suggest that a speed-accuracy tradeoff may be operating across this transition, but the lack of longitudinal studies makes this transition difficult to understand. The current study explored the longitudinal and bidirectional associations between accuracy and RT on the DCCS (mixed block) at 5, 6, and 7 years of age using cross-lagged panel analyses. The study also examined the roles of working memory and language, as potential longitudinal mediators between RT at Time *X* and accuracy at Time *X* + 1, and explored the role of inhibitory control. The sample consisted of 425 children from the Quebec Longitudinal Study of Child Development. Results show lagged associations from slower RT to greater improvements in accuracy between 5 and 6 years and between 6 and 7 years. Further, higher accuracy at 6 years predicted faster RT at 7 years. Only working memory acted as a partial mediator between RT at 5 years and accuracy at 6 years. These results provide needed longitudinal evidence to support theoretical claims that slower RT precedes improved accuracy in the development of cognitive flexibility, that working memory may be involved in the early stage of this process, and that accuracy and reaction time become more efficient in later stages of this process.

Keywords: Cognitive Flexibility, DCCS, Cross-Lagged Panel, Working Memory, Longitudinal, School Transition.

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Executive function (EF) skills are a set of cognitive skills that support the intentional, top-down control of thought, action, and emotion, and as such, they are required for goal-directed problem solving and adaptation to new circumstances (Miyake et al., 2000). Children's EF skills are positively related to their academic achievement both concurrently and prospectively, and children who struggle with EF skills are at risk of difficulties in school (Purpura, Schmitt, & Ganley, 2017; Willoughby, Wylie, & Little, 2019). EF skills are typically measured behaviorally as three skills: inhibitory control, working memory, and cognitive flexibility (Miyake et al., 2000). Inhibitory control involves deliberately suppressing an automatic response in favor of a less reflexive response, working memory involves maintaining and manipulating information in mind, and cognitive flexibility involves switching between perspectives or task sets. EF skills overlap in early childhood and become more distinct over time (Karr et al., 2018).

To assess cognitive flexibility in both adults and children, researchers typically use some type of task-switching paradigm. These paradigms can involve three types of trials: pre-switch trials or *baseline* trials, in which a single set of rules is required (e.g., sort by color); post-switch or *switch* trials, in which a different, incompatible set of rules is introduced; and *mixed* trials, in which participants must switch between the two rule sets (i.e., task sets) from trial to trial. Furthermore, within the mixed block, one set of rules might be required more often (i.e., dominant trials vs. non-dominant trials). Using accuracy or reaction time, depending on age, participants tend to perform best on pre-switch trials and worst on mixed block trials, with intermediate performance on post-switch trials. For children around 5 to 7 years of age, accuracy and/or reaction time on the mixed block provide the most sensitive measures of cognitive flexibility.

One of the most widely used measures of task switching in young children is the Dimensional Change Card Sort (DCCS, Zelazo, 2006). In the DCCS, children are shown two target stimuli (e.g., a blue rabbit and a red boat) and told to sort a series of mismatching bidimensional stimuli (e.g., red rabbits and blue boats) by matching them to one of the target cards. During the pre-switch block, they are told to sort by one dimension (e.g., color). Then, during the post-switch block, children are told to sort the same stimuli by the other dimension (e.g., shape), requiring a reversal of responses. Finally, during the mixed block, the instructed dimension varies unpredictably, from trial to trial as indicated by the presence or absence of a black dot. To switch between dimensions, children need to detect a change in the context (black dot or the absence of it) and select the dimension (rule) associated with that context.

Most 3-year-olds tend to *perseverate* on the post-switch trials of the DCCS, continuing to use the pre-switch rules, whereas 5-year-olds typically shift successfully. While accuracy on the mixed block continues to improve beyond age 5 years, it soon approaches ceiling, and subsequent age-related improvements in cognitive flexibility are more easily observed by measuring the reaction times (RT) associated with switching between rules (Zelazo, 2006). In short, switching difficulties (i.e., *switch costs*) can be measured either in terms of accuracy or in terms of RT, as is the case for measures of task switching more generally (e.g., Dots task in Davidson, Amso, Anderson, & Diamond, 2006).

Of particular interest, sometime around 6 years of age, children appear to make a speed-accuracy tradeoff, slowing down on difficult trials in order to maintain accuracy (Davidson et al. 2006; Zelazo et al., 2013). That is, before 6 years, children respond quickly on difficult trials but with a cost to accuracy (i.e., they are more likely to err). Around 6 years, children's accuracy typically improves but with a notable cost to RT, such that they slow down on difficult trials (Camerota, Willoughby, Magnus, & Blair, 2020; Tikhomirov, 1978). Subsequently, accuracy typically reaches ceiling and RT costs decrease (Davidson et al., 2006).

This type of developmental pattern, from low accuracy/fast RT to medium accuracy/slow RT to high accuracy/fast RT, is expected based on at least two complementary theoretical models, the conflict cusp model (van Bers, Visser, van Schijndel, Mandell, & Raijmakers, 2011) and the Iterative Reprocessing (IR) model (Zelazo, 2015). On the one hand, catastrophe theory, generally, and the conflict cusp model (van Bers et al., 2011), specifically, focuses on quantitative and qualitative characterizations of transitions in performance to explain the developmental pattern on the DCCS from low accuracy/fast RT to medium accuracy/slow RT to high accuracy/fast RT. Relying on catastrophe theory, van Bers et al. (2011) characterize development as occurring in a dynamic system open to different influences that are complex, hierarchical, and self-organized, and that produce different types of "catastrophic" change (or catastrophes) in the system (Thom, 1975; White, 1965). The cusp catastrophe, for example, involves a transition from one stable state to another. According to the conflict cusp model of the DCCS (van Bers et al., 2011), the sequence of change around age 6 years is discontinuous, and involves three consecutive phases: *perseveration* (i.e., stable state when children fail to switch), *transition* (i.e., when children slow down and use a mix of perseveration and switching strategies), and *switching* (i.e., stable state when children switch correctly between dimensions). According to this model, the first phase is marked by a high activation of the pre-switch rule, the last phase by the high activation of the post-switch rule, and the transition phase is an activation of both rules (pre-switch and post-switch). Catastrophe "flags" are used as criteria to determine or to predict the presence of a transition phase. These include, for example, sudden jumps (spurts in development), hysteresis (regressions in development), and critical slowing down (Van der Maas, & Molenaar, 1992).

Cross-sectional studies of children's performance across the ages of 3 and 6 years on the DCCS (van Bers et al., 2011; Dauvier, Chevalier, & Blaye, 2012; Mathy, Friedman, Courenq, Laurent, & Millot, 2015) have examined the conflict cusp model, using either latent Markov models (van Bers et al. 2011) or a generalized linear model (GLM) with latent classes (Dauvier et al. 2012) to identify participants in different subgroups corresponding to the three developmental phases of DCCS performance described above. Subgroups were consistently perseverative, transitional (using two different strategies, perseveration and set-shifting), and set-shifting, in terms of accuracy. Children in the set-shifting subgroups tended to be older than the children in the perseverative groups. However, the findings for RT were less clear. For example, Dauvier et al. (2012) found evidence of a speed-accuracy trade-off, but only for the set-shifting subgroup, not the transition sub-group, and van Bers et al. (2011) did not find significant RT differences among any of the subgroups although differences were in the correct direction. In sum, the conflict cusp model can account for the sequence of cross-sectional age group differences in accuracy on the DCCS (perseveration, transition, set-switching), but how these changes relate to changes in RT remains unclear. As such, the conflict cusp model does not really address the underlying neurocognitive mechanisms involved in these developmental transitions.

On the other hand, the IR model (Zelazo, 2015) does identify underlying neurocognitive mechanisms by which this developmental pattern might emerge. According to the IR model (Zelazo, 2015), the development of cognitive flexibility, and EF skills more generally, results from increases in the efficiency with which children iteratively reprocess information, or reflect upon, the problems they face. This reflection is triggered by the detection of conflict, uncertainty, or change, and it entails slowing down, which then allows for the (re-)formulation of complex explicit rules (using language) and the maintenance of this information in mind (i.e., working memory) while solving the problem. Cognitive flexibility is made possible by reflection and the formulation and use of a high-order rule for switching between rules. Although inhibitory control can facilitate reflection by blocking the influence of bottom-up tendencies, it is not strictly necessary for reflection (Zelazo, 2015); the inhibition of inappropriate responses can occur as a by-product of flexible rule use (requiring working memory and cognitive flexibility).

According to the IR model, the development of these skills in early childhood follows a particular sequence, which produces the developmental pattern seen in performance on the DCCS. Initially, young children fail to detect uncertainty or change and respond relatively quickly making them likely to err (low accuracy/fast RT). Around 5 years of age, most children can detect uncertainty, leading them to reflect on the situation, but they do so inefficiently at first. Consequently, while accuracy improves, responding is slower (medium accuracy/slow RT). With age, the efficiency (speed) of reflection increases even as accuracy plateaus near ceiling (high accuracy/fast RT). On this account, developmental increases in performance on the DCCS (cognitive flexibility) are made possible by increases in the likelihood and efficiency of reflection, together with the development of language and working memory.

Longitudinal studies using cross-lagged panel analyses could clarify whether the 3-phase sequence of low accuracy/fast RT to medium accuracy/slow RT to high accuracy/fast RT occurs *within* individual children. For example, does better accuracy at 6 years depend on more slowing earlier at 5 years (e.g., because children who detect conflict begin to slow down and reflect before responding)? Does the relation between accuracy and RT change from being positive to negative (i.e., from better accuracy and slower RT to better accuracy and faster RT), and if so, around what age does this tipping point typically occur?

In short, to understand changes in cognitive flexibility in this age range, it is important to examine both accuracy and RT scores and to do so at several different ages in the same children. The absence of longitudinal studies examining sequences of change in the relation between accuracy and RT makes it impossible to determine how these changes work together across time.

The current study explored the longitudinal and bidirectional associations between accuracy and RT scores on the DCCS mixed block, from preschool (5 years) to school age (6 and 7 years) using cross-lagged panel analyses (Figure 1). In order to clarify the roles of language and working memory, in the development of cognitive flexibility, the study examined vocabulary, measured by the Peabody Picture Vocabulary Test-Revised (PPVT-R: Dunn & Dunn 1981), and working memory, measured using the Random Object Span Task (ROST; Hongwanishkul, Happaney, Lee, & Zelazo, 2005), as potential mediators between DCCS RT and accuracy at Time *X* and DCCS RT and accuracy at Time *X* + 1, as proposed by the IR model (Zelazo, 2015). To clarify the role of inhibitory control in the development of flexibility, inhibitory control, measured by the Day-Night Stroop task (Gerstad, Hong, & Diamond, 1994) at 5 years, was examined as a potential mediator or moderator of the association between RT at 5 years and accuracy at 6 years.

Three complementary predictions were tested. First, in a transactional model, slowing down should be observed before reaching high accuracy/fast RT. That is, based on the conflict cusp and IR models, it was hypothesized that slower RTs at Time *X* would predict higher accuracy prospectively at Time *X* + 1. However, after children achieve a certain level of accuracy, individual differences in accuracy should then predict faster RT in the subsequent year. Specifically, higher accuracy at age 6 years should predict faster RT at age 7 years. Second, language and working memory were expected to mediate cross-lagged relations between RT and accuracy. Third, the indirect link between slowing down at 5 years and better accuracy at 6 years through inhibitory control—measured only at 5 years, should not be significant. Nevertheless, according to the IR model, inhibitory control could potentially moderate the link between RT and accuracy by blocking the influence of bottom-up tendencies and thus facilitating reflection.

Method

Participants

Participants were part of the first cohort of the QLSCD study (Jetté, Desrosiers, & Tremblay, 1997), one of the three cohorts taking part in the larger Quebec Longitudinal Project of Cognitive and Behavior Development, together with the second cohort of the QLSCD and the Quebec Newborn Twin Study (Geoffroy et al., 2010; Renouf et al., 2010; Séguin, Parent, Tremblay, & Zelazo, 2009). One thousand French-speaking or English-speaking families from urban areas (the Greater Montreal and Quebec City areas) and varied socioeconomic backgrounds were randomly selected from the Québec birth registry in 1996 and approached to participate. Of these, a total of 572 children (51.2% girls) were followed yearly from 5 months onwards (Jetté et al., 1997). Due to attrition and variations in participation rates at individual time points, the number of participants available for analyses at one time point at least was 425. However, the number available at each time point was necessarily lower, and decreased between time points (n = 386 at 5 years; n = 316 at 6 years; n = 276 at 7 years). At 6 years of age, 62% children were monolingual speakers of either French or English, 31% children spoke more than 1 language, and 6% children spoke more than 2 languages (MacLeod et al., 2019). All measures were administered in either French (> 90%) or English (< 10%) based on parent reports of which of these two languages was their child's preferred and/or most proficient. Language of administration on any given year was unrelated to test performance. Written informed consent was provided by parents and renewed at each follow-up.

Children were assessed in their homes by trained research assistants. Age 5 testing took place in the summer before children's entrance to kindergarten.

Measures

The Peabody Picture Vocabulary Test-Revised (PPVT-R)

The PPVT-R (Dunn & Dunn, 1981) or its French adaptation, the *Évaluation du Vocabulaire en Images Peabody* (Dunn, Thériault-Whalen, & Dunn, 1993) were used to assess receptive vocabulary at 5, 6, and 7 years in English- and French-speaking participants, respectively. The examiner presented a series of trials each depicting four pictures. For each trial, children identified the picture that represented the word said by the examiner. Both tests contained five training items, followed by a subset of the total items that were presented to children based on their age and abilities following a standard procedure to establish floor and ceiling items. The normative population for the EVIP was drawn from French speakers living in Canada. Scores were standardized within each language version of the test and combined into a single receptive vocabulary index.

The Random Object Span Task (ROST)

The ROST (Hongwanishkul, Happaney, Lee, & Zelazo, 2005) is an adapted version for children of the Self-Ordered Pointing Task (Petrides & Milner, 1982), which assesses visual working memory. In this version, images were presented to the children on a laptop computer fitted with a touch screen (See Supplemental Figure S1 for an example of a trial in the Supplemental Materials). Children were given the following instruction at the beginning of the task, "You will see pictures on the screen. You have to choose all of them, but only one at a time. When you have selected one with your finger, the pictures will get mixed on the screen. Then you select a different one. Do you understand? Let's start." To perform correctly on each trial, children needed to keep in mind and update the images already selected so as not to choose the same images again. There were two trials per level. The number of images to be remembered increased by 1 image for each increasing level with a maximum of 9 images. The score consisted of the total number of correct trials.

The Day-Night Stroop Task

The Day-Night Stroop task (Gerstadt, Hong, & Diamond, 1994; Montgomery & Koeltzow, 2010) was available only at 5 years[1]. The task included two parts. In the first part, the experimenter showed a picture of the sun or the moon on a computer screen. In the first part, for 10 trials, participants were first instructed to say "day" when shown the sun and "night" when shown the moon (congruent trials) In the second part, for 10 trials, participants were instructed to reverse that response set by saying "night" to a picture of a sun and "day" to a picture of a moon (incongruent trials). No feedback was provided on any trial. Accuracy consisted of the number of correct responses of the 10 trials in the second part of the task.

The Dimensional Change Card Sorting task (DCCS)

The mixed blocks version of the DCCS was administered at 5, 6, and 7 years on a laptop computer fitted with a touch screen. In the current version of the task (adapted from Zelazo, 2006), participants sorted test cards depicting bidimensional pictures (i.e., orange horses and brown cups) that could be sorted by either one or the other dimension (i.e., color or shape) onto two targets that

[1] Children approached ceiling performance on Day-Night Stroop at 5 years, so we do not have repeated measures for the Day-Night Stroop for all 3 time points as we do for the DCCS, PPVT, and the ROST.

depicted an orange cup or a brown horse. In other words, either target card could be pressed depending on the sorting rules (e.g., a brown horse could be sorted with the orange horse or the brown cup). The two sorting rules changed arbitrarily on a quasi-random basis across 20 trials, as indicated by the presence or absence of a black dot. The children were given the following instructions at the beginning of the task and again after 10 trials: "In the next game, when there is a black dot like this one, we are playing the ____ (e.g., color) game. So all the ____ (e.g., orange) pictures go here and all the ____ (e.g., brown) pictures go there. If it is ____ (e.g., orange) it goes there and if it is ____ (e.g., brown) it goes there. However, when there is no black dot we are playing the ____ (alternate; e.g., shape) game. So all…." These instructions were followed by two demonstrations and four practice trials with feedback. The DCCS score was based on 20 trials, which included 15 trials according to one dimension (i.e., the dominant dimension) and 5 trials according to the other (i.e., the non-dominant dimension). Before each trial, short verbal instructions were given to remind children of the sorting rules ("Remember, when there is a black dot you are playing the color game and when there is no black dot you are playing the shape game."). No feedback was provided on the test trials.

Accuracy consisted of the number of correct responses in the mixed block, and RT was defined as the median response time across the 20 trials of the mixed block. RTs greater than 10.00 s were coded as 10.00 s. Two median RT scores were computed, one for correct trials only, and one for all trials (both correct and incorrect). Outliers more than 3 *SD* from the mean and below 0.10 s were discarded. Also, participants' DCCS scores were excluded in a given year if participants were not cooperative, there was a problem in the administration, or the environment was disruptive, or for whom child state ratings during administration were missing. A total of 36 DCCS scores at 5 years, 15 scores at 6 years, and 5 scores at 7 years were excluded for those reasons.

Although RT scores are often based on each participants' median RT for correct trials only, we used RTs from both incorrect and correct trials for several reasons. First, the mean number of correct trials at 5 years (*M* = 12.41, *SD* = 3.75, range 2-20 20 trials, see Table 1) is typically smaller than at later ages, which reduces the reliability of the RT score. Second, the number of trials used for RT calculations varied as a function of participants' accuracy, which created a covariance between the number of trials and accuracy.[2]

Analysis Plan

The first objective of this study was to examine the transactional link between RT and accuracy scores on the DCCS mixed Block at 5, 6, and 7 years of age, as predicted by the IR model (Zelazo, 2015) and the conflict cusp model (van Bers et al., 2011). In other words, does the total accuracy score at Time *X* predict RT at Time *X* + 1 and vice versa (see Figure 1)? The second objective was to test for possible mediator roles of language and working memory at age 5 and 6 in the development of subsequent cognitive flexibility (at age 6 and 7, respectively), as predicted by the IR model—that is, whether there is an indirect effect between RT at Time *X* and accuracy at Time *X* + 1 via PPVT-R scores (i.e., receptive language as indexed by vocabulary) at Time *X* and between RT at Time *X* and accuracy at Time *X* + 1 via the ROST score (i.e., working memory) at Time *X*. Indirect effects included the cross-sectional association between RT and potential concurrent mediators (PPVT-R and ROST; i.e. both measured at the same time point (Time *X*))

[2] See Supplemental Materials Figures S2a and S2b for analyses conducted with median RTs for correct trials only. When comparing the results of each analysis (one with RT scores based on all trials and the other with RT scores based on correct trials only), lower *p* values were found for similar betas in the model based on both correct and incorrect trials presented in the Results section, suggesting that the estimate of measurement error was reduced with this method.

rather than longitudinal (cross-lag) associations between RT and mediators. Cross-sectional associations were favored as it was hypothesized that reflection co-occurs with the development of language and working memory. The third objective was to test whether inhibitory control at age 5 (the time point at which it was available) played a mediating or moderating role in the association between reaction time at age 5 years and accuracy at age 6 years.

To study this developmental process across 5, 6 and, 7 years old, cross-lagged models were performed with structural equation modeling using Mplus Version 7 (Muthén & Muthén, 2012, Asendorpf, 2021).[3] Maximum likelihood with robust standard errors estimator was used to examine cross-lagged and autoregressive effects. Cross-lagged analyses allow testing for transactional associations between variables across time (e.g., RT at 5 years associated with accuracy at 6 years), while controlling for their stability. We used a combination of indices to evaluate the adequacy of the fit of the model: Chi-square, Tucker-Lewis, the "comparative fit index" (CFI) at 0.90 or more, and the root mean square error of approximation (RMSEA) at 0.08 or less (Little, 2013). Indirect effects were tested using the bias-corrected bootstrap approach, which is recommended for small samples, as it demonstrates increased power and reliability in smaller samples (Mackinnon,

[3] Cross-lagged panel model (CLPM) was chosen rather than the Random Intercept Cross-Lagged panel model (RI-CLPM), because CLPM are preferred when the main goal is to assess the potential effects of change-related causes that make persons different from other persons—and not from the (long-term) average level—(Asendorpf, 2021). RI-CLPM is better suited for studies of states that typically use shorter time lags (e.g., days) between assessments and are not concerned about systematic long-term changes like here (e.g. yearly assessments).

Lockwood, & Williams, 2004; Fritz & MacKinnon, 2007). In order to test the main effect of DCCS RT, and those in interaction with inhibitory control at 5 years on DCCS accuracy at age 6 years, a series of linear regressions were conductied using Mplus Version 7 (Muthén & Muthén, 2012). We tested models of the moderating role of inhibitory control at age 5 years in the pathways from DCCS RT and accuracy, including both working memory and language at 5 years. Predictors and moderators were standardized, and full information maximum likelihood (FIML) was used for handling missing data. Predictors and moderators were standardized, and full information maximum likelihood (FIML) was used for handling missing data. Repeated measures ANOVAs with Greenhouse-Geiser correction were used to describe age effects for accuracy, RT, PPVT-R, and ROST scores, using SPSS Version 26.

Results

Descriptive Statistics

Table 1 describes age-related scores on three measures at each age: DCCS (accuracy and RT), PPVT-R and ROST, and the Day-Night Stroop at age 5 years. There were linear age changes for DCCS accuracy ($F(1.91, 35.75) = 121.69$, $p < .001$, $\eta_p^2 = .40$), PPVT-R ($F(1.84, 381.91) = 907.67$, $p < .001$, $\eta_p^2 = .81$), and ROST scores ($F(1.97, 438.02) = 117.20$, $p < .001$, $\eta_p^2 = .35$), suggesting that these increased linearly with age. In contrast, for DCCS RT, the age effect was quadratic ($F(1.85, 355.85) = 3.35$, $p = .04$, $\eta_p^2 = .03$), suggesting that age-related changes were not linear. Notably, although the mean increase in DCCS RT was not significant between 5 and 6 years of age ($p = .45$), DCCS RT was significantly faster at 7 compared to 6 years of age ($p = .01$).

As shown in Table 2, accuracy and RT on the DCCS were not correlated concurrently at 5, 6, or 7 years. However, significant prospective associations between DCCS RT and accuracy were found: RT at 5 years was positively and prospectively correlated with accuracy at 6 years and RT at 6 years was correlated with accuracy at 7 years. DCCS accuracy was significantly associated with PPVT-R concurrently and prospectively across all ages. In contrast, DCCS RT did not correlate with PPVT-R concurrently at any time point, but PPVT-R at 5 years was positively and prospectively associated with DCCS RT at 6 years and DCCS RT at 6 years was positively associated with PPVT-R at 7 years. There were no significant concurrent correlations between DCCS RT and PPVT-R at ages 5 and 6 years. DCCS accuracy was concurrently and positively associated with the ROST at 6 and 7 years, and ROST at 5 years was prospectively associated with DCCS accuracy at 6 years and vice versa (i.e. bidirectional effects from 5 years to 6 years). DCCS RT scores at 5 and 7 years, but not 6 years, were concurrently and positively correlated with the ROST. Prospective associations between DCCS RT and ROST were also found: the ROST at age 7 was positively related to RT at all ages. Thus, slower RTs at all years were predictive of better working memory at age 7 years. Finally, and of note, although the Day-Night Stroop at 5 years was significantly and prospectively correlated with DCCS accuracy at 6 and 7 years, with all PPVT scores, and concurrently with ROST at age 5 years, it was not significantly associated with any DCCS RT scores nor with DCCS accuracy at 5 years.

Cross-lagged Models

A first cross-lagged model examining accuracy and RT at 5, 6, and 7 years (Figure 2) provided an acceptable model fit: χ^2 = 2.18 (*df* = 3), RMSEA = 0.00, CFI = 1.00, SRMR = 0.018. Autocorrelations show that accuracy and RT were relatively stable across time (all $\beta \geq 0.15$). More interestingly, three significant cross-lagged regression paths were found: slower RT at 5 year was significantly associated with improvement in accuracy at 6 years (i.e., change in accuracy from age 5 to 6 years; $\beta = 0.16$, $p = .01$). The same path was found between 6 and 7 years ($\beta = 0.19$, $p = .001$). In contrast, better accuracy at 6 years was negatively associated with changes in RT at 7 years ($\beta = -0.13$, $p = .04$). Accuracy and RT at 7 years were also negatively associated, but not significantly ($\beta = -0.09$, $p = .16$). These results indicate that as more children reached a threshold on accuracy at around 6 years (*M* = 15.5/20, *SD* = 4.1), the direction of the association between accuracy at Time *X* and RT at Time *X* + 1 becomes reliably negative. No other cross-lagged or cross-sectional associations were found.

A second cross-lagged model of accuracy and RT using the ROST at 5, 6, and 7 years (Figure 3) tested the indirect effect hypothesis. The model had an acceptable fit: χ^2 = 16.145 (*df* = 11), RMSEA = 0.033, CFI = .943, SRMR = 0.037. Only one significant indirect effect was found from DCCS

RT at 5 years to DCCS accuracy at 6 years through ROST accuracy at 5 years (*ab* = .027, Bootstrap 95% CI: .009 – .053), suggesting that the association between lower RT at 5 years and higher accuracy at 6 years was partly explained by working memory at 5 years. However, the direct cross-lagged path from RT at 5 years and accuracy at 6 years remained significant ($\beta = 0.13$, *p* =.031), indicating that working memory at 5 years explained 19% of the cross-lagged path from lower RT at 5 years to higher accuracy at 6 years. The indirect effect between DCCS RT at 6 years to DCCS accuracy at 7 years through ROST accuracy at 6 years was not significant (*ab* =.005; *p* = .62). Indeed, the direct cross-lag effect from RT at 6 years and accuracy at 7 years did not change with the inclusion of ROST to the model. Cross-lagged models with language or inhibitory control as the mediator were not conducted because the absence of concurrent correlations between DCCS RT and PPVT-R at ages 5 and 6 years or DCCS RT and Day-Night Stroop Task at age 5 years precluded their use as potential mediators. Finally, in order to test whether the association between DCCS RT at age 5 years and later accuracy at age 6 years varied as a function of inhibitory control at age 5 years, inhibitory control was entered into the models as a moderator (by creating interaction terms between inhibitory control and RT, as well as inhibitory control and mediators (WM and language). Results showed that inhibitory control did not moderate the associations between RT and accuracy on the DCCS ($\beta = -0.04$, *p* =.61), or the indirect effects from RT to accuracy through working memory or language (significance of interaction terms were all above p>0.23).

Discussion

Although many cross-sectional studies have examined the development of cognitive flexibility using accuracy with preschoolers complemented later by RT at school age, there are no previous longitudinal studies examining how both indicators are related across time during this rapid period of development. To fill this gap, we examined developmental changes in the relation between both indicators using the DCCS (mixed block) across 5, 6 and 7 years using a general cross-lagged approach. This approach allowed us to test the predictive power of each indicator from one time to the next. Also, the study examined the potential roles of working memory and vocabulary as potential mediators of this developmental process exploring the role of inhibitory control at 5 years of age.

As expected, and consistent with Cepeda, Kramer, and Sather (2001) and others, there were age-related improvements in both accuracy and RT, although accuracy and RT were not related within years. However, cross-lagged panel analyses showed positive lagged associations from RT to accuracy (i.e., longer/slower RTs predicted later higher accuracy) between 5 and 6 years, and between 6 and 7 years. Additionally, as predicted, working memory, measured by the ROST, acted as a partial mediator between RT at 5 years and improvements in accuracy at 6 years, supporting the notion that slowing down and reflecting may be reliant on working memory. In contrast, and contrary to expectations, receptive vocabulary was not associated with RT, precluding assessing its role as a potential mediator between RT and accuracy over time. Similarly, as predicted, the association between RT and accuracy was not mediated by inhibitory control, as measured by the Day-Night Stroop. In contrast with our prediction, the association between RT and later accuracy did not vary as a function of level of inhibitory control.

These cross-lagged results provide longitudinal evidence about the development of cognitive flexibility, showing multi-year shifts in speed-accuracy trade-offs. The current results are consistent with both the conflict cusp and IR models. The conflict cusp model suggests a 3-phase sequence of developmental changes in the relation between accuracy and RT: (1) low accuracy/fast RT; (2) medium accuracy/slow RT; (3) high accuracy/fast RT. This sequence results from a transition from one stable state of a dynamic system to another, via a relatively rapid transition phase. Consistent with this model, we found that the youngest children had the lowest accuracy and that accuracy increased significantly across age. Although RTs did not differ significantly between 5 years and 6 years, they did between 6 and 7 years old: A change from slower to quicker responses was found between 6 and 7 years old. The lack of statistical significance in the RT mean differences across 5 and 6 years may be due to the particular age range included in the current study. Children younger than age 5 years might be more likely to make impulsive quick responses (Diamond, Carlson, & Beck, 2005). Further, the conflict cusp model also suggests that children within a given age may be at different developmental phases, and this heterogeneity in RT may be masked when comparing overall yearly means. This may account for why accuracy and RT parameters were not concurrently significantly associated with each other.

The cross-lagged associations observed in this study are also consistent with the IR model. Slowing down at age 5 or age 6 years predicted improved accuracy the following year, indicative of an increase in reflection preceding an increase in proficiency. However, as accuracy improved at age 6, it became associated with later speeding up between 6 and 7 years, consistent with an increase in the efficiency of reflection over time. Indeed, slower reaction times at Time *X* predicted improved accuracy between Times *X* and *X* + 1, and working memory at 5 years partially explained the link between slowing at 5 years and improvements in accuracy at 6 years. Although slowing down at ages 5, 6, and 7 years was related to working memory at age 7 years, there was no evidence of working memory mediating the association between slowing at age 6 years and improvements in accuracy at age 7 years. In other words, this finding is consistent with the suggestion that slowing down, reflecting and manipulating information in mind, helps children to create a complex system of rules, which allows them to behave in accordance with changing conditions (i.e., color rule with border, shape rule without border), and that working memory is needed early in this process. Once established, however, additional improvements in working memory no longer benefit later improved efficiency. Finally, the tendency for the cross-sectional beta statistic to become negative and approach significance at 7 years and the negative relation between improvement in accuracy at 6 years and improvement (faster) reaction time at 7 years, together suggest that this reflective reprocessing becomes more efficient and less costly on RT around 7 years.

Consistent with the IR model, our results support the idea that a shift from a reliance on more bottom-up influences (low accuracy/faster RT) through a transitional period (medium accuracy/slower RT) to the use of a more top-down regulation strategy (high accuracy/faster RT) happens somewhere between 5 to 7 years of age for most children on the DCCS. A shift around this period has also been found for other, related cognitive control skills, such as for reactive and proactive control (Chevalier & Blaye, 2009). Indeed, children tend to shift from using reactive to proactive control after about 5 years of age, and this transition also appears to rely on working memory (Chevalier, Martis, Curran, & Munakata, 2015; Van der Maas & Molenaar, 1992).

Limitations and Future Directions

The data described here are based on a single measure of cognitive flexibility, and on children between the ages of 5 and 7 years. Future longitudinal research using additional measures of cognitive flexibility, and examining performance across a wider age range (e.g., from 3 to 9 years) could allow for a clearer picture of accuracy and RT developments, as well as a better understanding of what might be underlying these changes.

Although we found that as predicted, working memory partially mediated the relation between RT and accuracy, receptive vocabulary did not concurrently correlate with RT, which prevented assessing it as a potential mediator between RT and accuracy over time. It is possible that receptive vocabulary is not sensitive enough as an index of the underlying language skills required for representing the (re-)formulation of complex explicit rules. A more extensive assessment of children's linguistic skills might reveal a mediating role for language.

Similarly, inhibitory control, as measured by the Day-Night Stroop, did not concurrently correlate with either of the DCCS indicators. Unfortunately, because of ceiling effects, inhibitory control was not assessed longitudinally at 6 or 7 years, so we cannot determine whether it may have had a mediator role later in the process. Further studies with other measures and a larger age range could help clarify at what ages, if any, inhibitory control is associated with accuracy and slower RT on measures of cognitive flexibility.

Further, many factors in the social environment of children may also be at play, whether at home, with peers, or at school, and particularly during the school transition. A study of such factors could help identify under which conditions cognitive flexibility is facilitated or hindered and inform prevention science.

One limitation of the cross-lagged panel-model approach taken in the current study is that it is a variable-centered approach that can confound within-person processes and more stable between-person differences (Hamaker, Kuiper, & Grasman, 2015). As DCCS studies have found variability within and between age groups (Camerota, Willouogby, & Blair, 2019), it would be useful to design future studies with shorter time lags between assessments (e.g., assessments twice a year) to disaggregate between- and within-person variation using RI-CLPM (Willoughby, Wylie, & Little, 2019). Models suggest that the sequence of developmental changes in the relation between accuracy and RT (i.e., (1) low accuracy/fast RT; (2) medium accuracy/slow RT; (3) high accuracy/slow RT) occurs within persons. Accordingly, for within-person variation in children who have not yet entered the transition, fast RT will be associated with poor accuracy; for those just entering the transition, slow RT will be associated with better accuracy; and for those who have more fully transitioned, high accuracy will be predicted by fast RT.

Exploring inter-individual differences would also help to clarify methods for combining accuracy and RT scores, and perhaps help identify important variables responsible for initiating developmental transitions. Although methods for combining accuracy and RT scores are based on the idea that beyond some age, speed reflects cognitive flexibility better than accuracy alone, the complex relations between RT and accuracy found in the current study warrants caution with this approach. Moving ahead may require revisiting and optimizing methods for combining accuracy and RT to clarify under which conditions slower RTs are also valuable indicators of developmental changes in cognitive flexibility during this specific period of development. Future research might also usefully employ hierarchical diffusion modeling to decompose performance into separate processing components (e.g. Manning, Wagenmakers, Norcia, Scerif, & Boehm, 2021; Ratcliff & McKoon, 2008; Vandekerckhove, Tuerlinckx, & Lee, 2011).

Conclusion

This study is the first to explore the longitudinal, bidirectional relations between accuracy and RT scores on the DCCS (mixed block) at 5, 6, and 7 years with cross-lagged panel. Results of the cross-lagged panel analyses showed positive lagged associations from RT scores to accuracy between 5 and 6 years and again, between 6 and 7 years, and negative lagged associations from accuracy to RT scores between 6 and 7 years. Working memory also partly mediated the early association between RTs and accuracy. This sequence of change suggests transition phases marked by a slower reaction time before an improvement in accuracy (e.g., from perseveration to switching), and then by improved accuracy preceding increased response efficiency. Being able to maintain the relevant rules in mind may be an early step for success, as proposed by the IR model. When it comes to developmental change, the naïve assumption is to believe that children simply become "better and faster" at what they do—be it in learning to read or learning to run—and that speed and accuracy develop in tandem. However, the results of the current study suggest that the relation between speed and accuracy may be more complex. As children improve in one, they appear to take a step back in the other before catching up again. Indeed, for some areas of development, "two steps forward, one (slower) step back" might be the norm.

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Table 1

Descriptive data at 5, 6 and 7 years old for all model variables

Measures	N	Min.	Max.	M.	SD.	Skewness	Kurtosis
1. ACC DCCS 5 y	348	2	20	12.41	3.75	-0.04	-0.32
2. ACC DCCS 6 y	301	3	20	15.50	4.10	-0.69	-0.45
3. ACC DCCS 7 y	269	5	20	17.49	3.41	-1.41	0.96
4. RT DCCS 5 y	348	0.88	11.59	3.57	1.42	1.17	3.12
5. RT DCCS 6 y	301	1.56	8.10	3.81	1.16	0.94	1.25
6. RT DCCS 7 y	269	1.21	8.24	3.49	1.00	0.88	2.08
7. PPVT-R 5 y	380	10	108	61.65	18.21	-0.06	-0.36
8. PPVT-R 6 y	308	22	130	85.82	16.09	-0.58	0.67
9. PPVT-R 7 y	272	58	141	101.19	14.36	-0.15	0.10
10. ROST 5 y	386	0	12	6.00	2.39	0.01	-0.20
11. ROST 6 y	316	1	13	7.35	2.28	-0.07	-0.48
12. ROST 7 y	276	1	16	9.19	2.49	-0.24	0.07
13. Day-Night 5 y	384	0	10	8.59	2.64	-2.22	3.99

Note. Due to attrition, loss at follow up, and variations in participation rates425 participants were available at for analyses at one time point at least. However, the number available at each time point was necessarily lower, and, Ns decreased from one time to the next.

ACC = Accuracy; RT = Reaction time; DCCS = Dimensional Change Card Sort; PPVT-R = Peabody Picture Vocabulary Test-Revised; ROST = Random Object Span Task; Day-Night = Day-night Stroop task.

Table 2

Bivariate correlation matrix for all model variables

Measures	1	2	3	4	5	6	7	8	9	10	11	12
1. ACC DCCS 5 y	-											
2. ACC DCCS 6 y	.14*	-										
3. ACC DCCS 7 y	.20**	.17	-									
4. RT DCCS 5 y	.04	.15*	.06	-								
5. RT DCCS 6 y	.02	.05	.17**	.15**	-							
6. RT DCCS 7 y	-.10	-.13	-.08	.03	.21**	-						
7. PPVT-R 5 y	.12*	.18**	.21**	.05	.14*	.01	-					
8. PPVT-R 6 y	.20**	.28**	.26**	.07	.09	-.01	.67**	-				
9. PPVT-R 7 y	.14*	.21**	.27**	.05	.18**	-.00	.61**	.71**	-			
10. ROST 5 y	.06	.16**	.05	.21**	.03	.06	.22**	.14*	.09	-		
11. ROST 6 y	.08	.22**	.11	.09	.10	.02	.14*	.13*	.24	.25**	-	
12. ROST 7 y	.01	.17*	.20*	.19**	.18**	.14 *	.23**	.20**	.25**	.21**	.19**	
13. Day-Night 5y	.09	.21**	.19**	.05	.08	-.03	.29**	.30**	.30**	.23**	.09	.09

*p < .05; **p < .01. ACC = Accuracy; RT = Reaction time; DCCS = Dimensional Change Card Sort; PPVT-R = Peabody Picture Vocabulary Test-Revised; ROST = Random Object Span Task; Day-Night = Day-Night Stroop Task.

Figure captions

Figure 1

Longitudinal transactional model of the relation between accuracy and reaction time on the DCCS at 5, 6 and 7 years old.

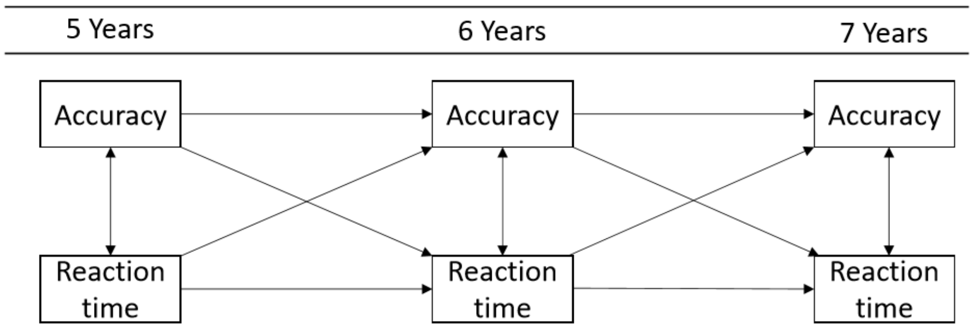

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Figure 2

The cross-lagged model trials showing the longitudinal associations between the DCCS Accuracy Scored and DCCS advanced Median Reaction Time on correct and incorrect trials from 5, 6 and 7 years old.

Note: Standardized coefficients β are shown; *p < .05; ** p < .01. Statistically significant paths are shown with solid lines, and nonsignificant paths are shown with dashed lines. Blue line = negative association. Red line = positive association. Estimation by MLR. Model fit: χ^2 = 2.180(df : 3), RMSEA = .000, CFI =1.00, SRMR = .018.

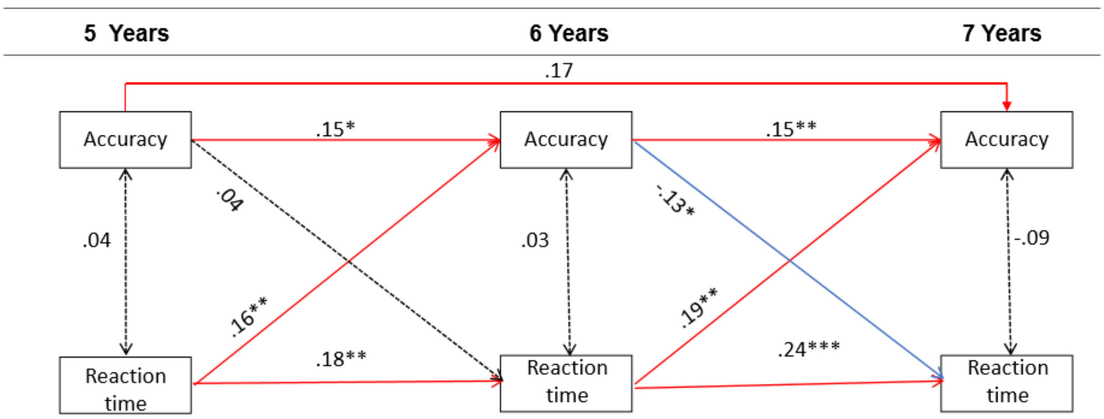

Figure 3

The cross-lagged model trials showing the longitudinal associations between the DCCS Accuracy Scored and DCCS advanced Median Reaction Time on correct and incorrect trials and ROST from 5, 6 and 7 years old.

Note: Standardized coefficients β are shown; *p < .05; ** p < .01. Statistically significant paths are shown with solid lines, and nonsignificant paths are shown with dashed lines. Blue line = negative association. Red line = positive association. Estimation by MLR. Model fit: χ^2 = 16.145 (df : 11), RMSEA = .033, CFI = .943, SRMR = .037.